

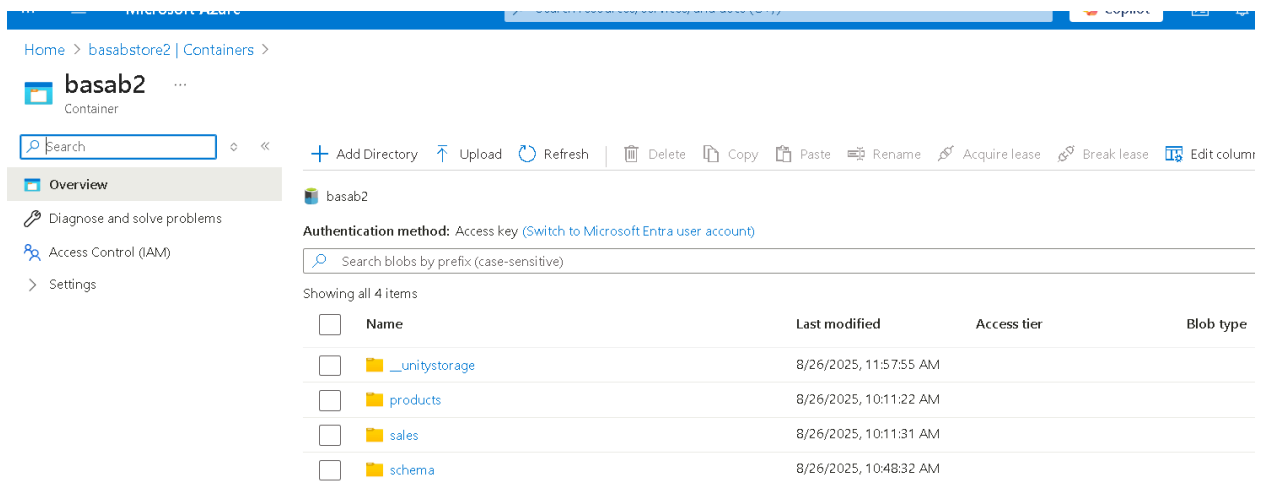
Basab's :Retail Compliance & Analytics Platform

Email :

Tredence LabsKraft4

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Storage Account Creation :



Home > basabstore2 | Containers >

basab2
Container

Search

Overview

- Diagnose and solve problems
- Access Control (IAM)
- Settings

basab2

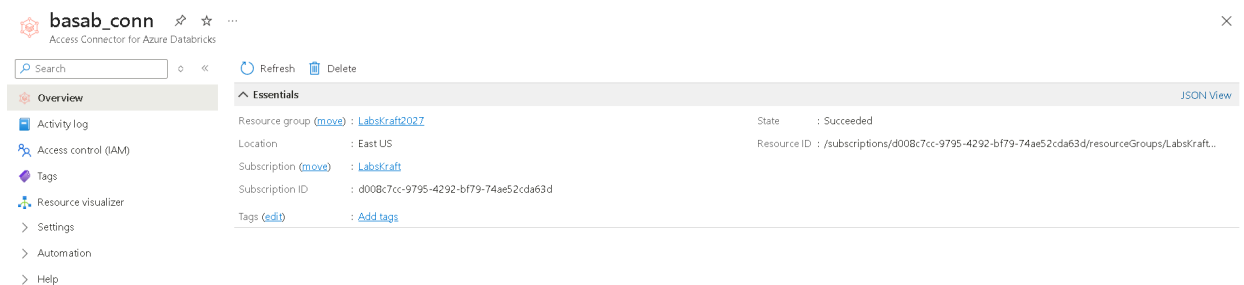
Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Search blobs by prefix (case-sensitive)

Showing all 4 items

☐	Name	Last modified	Access tier	Blob type
☐	__unitystorage	8/26/2025, 11:57:55 AM		
☐	products	8/26/2025, 10:11:22 AM		
☐	sales	8/26/2025, 10:11:31 AM		
☐	schema	8/26/2025, 10:48:32 AM		

Databricks Access Connector Creation :



basab_conn
Access Connector for Azure Databricks

Search Refresh Delete

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Settings
- Automation
- Help

Essentials

State : Succeeded

Resource ID : /subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/LabKraft...

Resource group (move) : [LabsKraft2027](#)

Location : East US

Subscription (move) : [LabsKraft](#)

Subscription ID : d008c7cc-9795-4292-bf79-74ae52cda63d

Tags (edit) : [Add tags](#)

[JSON View](#)

Unity Catalog Creation :

Catalog Explorer >

 **basab_catalog3**    Share ▾ Create schema

[Overview](#) [Details](#) [Permissions](#) [Workspaces](#)

Description

 AI generate Add

3 schemas

Name	Owner	Created at
 all_delta	tredence4@michaelkloudkraft...	Aug 26, 2025, 11:56 AM
 default	tredence4@michaelkloudkraft...	Aug 26, 2025, 10:17 AM
 information_schema	System user	Aug 26, 2025, 10:17 AM

About this catalog

Owner Tredence LabsKraft4 

Tags

DLT Code :Bronze DLT :

```

import dlt
@dlt.table(
    comment="Bronze delta table for staging"
)
def bronze_orders():
    source_path="abfss://basab2@basabstore2.dfs.core.windows.net/sales/"

    schema_location = "abfss://basab2@basabstore2.dfs.core.windows.net/schema/"

    raw_stream = (
        spark.readStream
        .format("cloudFiles")
        .option("cloudFiles.format", "csv") # Pass directly
        .option("cloudFiles.schemaLocation", schema_location)
        .load(source_path)
    )
    bronze_df=raw_stream
    bronze_df = (
        raw_stream
        .withColumn("sale_id", col("sale_id").cast("int"))
        .withColumn("product_id", col("product_id").cast("int"))
        .withColumn("quantity", col("quantity").cast("int"))
        .withColumn("sales_amount", col("sales_amount").cast("double"))
        .withColumn("sale_date", col("sale_date").cast("timestamp"))
    )
    return bronze_df

```

DLT Code :Silver DLT :

```

@dlt.table(
    comment="Silver delta table for transformations"
)
@dlt.expect("valid_quantity","quantity>0")
@dlt.expect("valid_sales_amount","sales_amount>0")
def silver_orders():
    bronze_df=dlt.read("bronze_orders")
    product_df=dlt.read("products_table")

    #Join Korlan 🤖
    silver_df = bronze_df.join(product_df, "product_id", "left") # join on product_id

    #New col : Discount ache ki nei ? (Extra Col)
    silver_df=silver_df.withColumn("has_discount",when( col("discount_rate")>0 ,lit(True) ).otherwise(lit(False)))
    # New Col : Ekta Unit er Price Koto ?
    silver_df=silver_df.withColumn("unit_price",when( col("quantity")>0 ,col("sales_amount")/col("quantity") ).otherwise(lit(0)))
    # New Col : Kerom Review hoache : high/medium/low
    silver_df=silver_df.withColumn("renewal_tier",when( col("sales_amount")>1000,"High" ).when(col("sales_amount")>500,"Medium").otherwise("Low"))

    #New Col : Koto Quantity Load royeche : "Bulk/Medium/Low"
    silver_df=silver_df.withColumn("quantity_load",when(col("quantity")>10,"Bulk" ).when(col("quantity")>5,"Medium").otherwise("Low"))
    #New Col : Extracting Store Number :
    silver_df=silver_df.withColumn("store_number",regexp_extract(col("store_id"), r"Store-(\d+)", 1).cast("int") )
    #New Col : Filling Nulls

```

```

#New Col : Filling Nulls
silver_df=silver_df.fillna(
    {
        "discount_rate":0,
        "unit_price":0,
    }
)
# Kokhon transform korchi tar timestamp add : New Col
silver_df = silver_df.withColumn("_ingest_ts", current_timestamp())
# Jodi duto Sales transaction er ID same hoi then it's invalid, so drop the duplicates
silver_df = silver_df.dropDuplicates(["sale_id"])

#Finally return the table
return silver_df

```

DLT CODE : GOLD CODE :

Daily Region Sales Gold Table :

```

@dlt.table(
    comment="Gold delta table for aggregations"
)
def daily_region_sales():
    silver_df=dlt.read("silver_orders")

    silver_df=silver_df.withColumn("sales_date_only",to_date(col("sale_date")))

    gold_agg1=silver_df.groupBy("region","sales_date_only").agg( sum("sales_amount").alias("total_sales") , count("sale_id").alias("total_orders"))

    return gold_agg1

```

Category Performance Gold Table :

```
@dlt.table(  
    comment="Gold delta table for agg2"  
)  
def category_performance():  
    silver_df=dlt.read("silver_orders")  
  
    gold_agg2=silver_df.groupby("category").agg( sum("sales_amount").alias("total_sales") , count("sale_id").alias("total_orders"))  
  
    return gold_agg2
```

Store Performance Gold Table :

```
@dlt.table(  
    comment="Gold delta table for agg3"  
)  
def store_performance():  
    silver_df=dlt.read("silver_orders")  
    gold_agg3=silver_df.groupby("store_number").agg( sum("sales_amount").alias("total_sales") , count("sale_id").alias("total_orders"))  
  
    return gold_agg3
```

Store Category Gold Table :

```
@dlt.table(  
    comment="Gold delta table for agg4"  
)  
def store_category_performance():  
    silver_df=dlt.read("silver_orders")  
    gold_agg4=silver_df.groupby("store_number","category").agg( sum("sales_amount").alias("total_sales") , count("sale_id").alias("total_orders"))  
  
    return gold_agg4
```

Full Pipeline Working :

Pipeline event log details

Summary

JSON

ERROR

2025-08-26 16:03:52 IST

update_progress

Update 89b510 has failed while trying to update table 'basab_catalog3.all_delta.silver_orders'.

Diagnose error

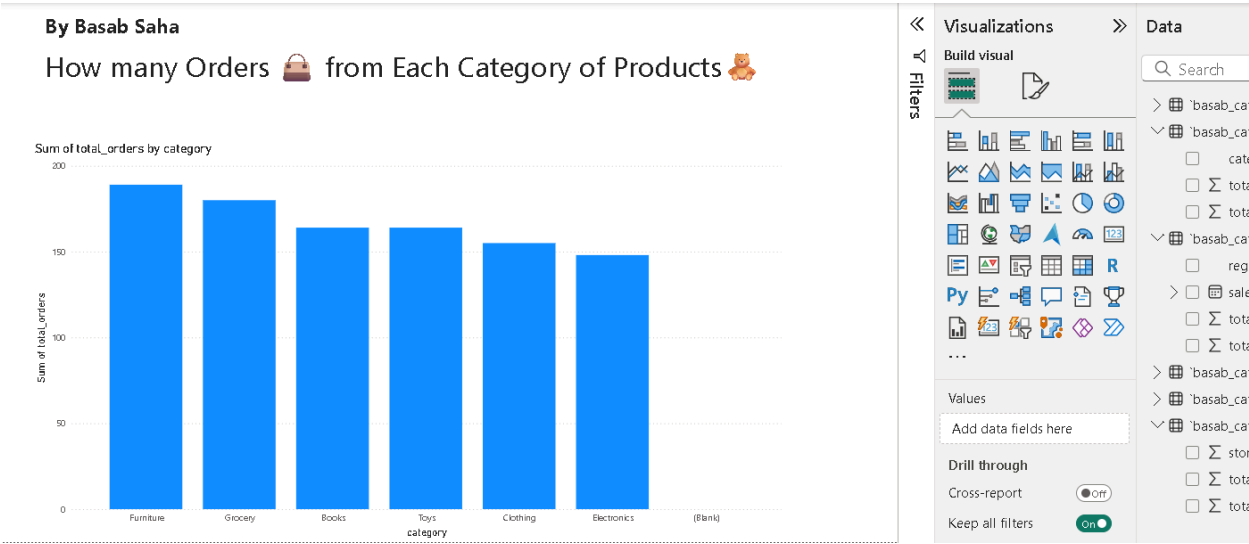
Error details

```

com.databricks.sql.transaction.tahoe.DeltaAnalysisException: [DELTA_INVALID_CHARACTERS_IN_COLUMN_NAMES] Found invalid character(s) among ' ,;{}()\n\t=' in the column names of your schema.
Invalid column names: `PK[Content_Types].xml (J@5Q>[i{2
hnm2R^7/ rZY@1_ f q R4D AJ h > V z Z 9 NV 8f { ^
-I "{ v ^ P!XS) bR r K s ( 3 ` c 0 0 0 0 0 0 0 0 7 M4 0 0 0 0 0 0 Z Ek + | \ z ( P 0 0 6 h _ - [ @ ! P k
2 n } ? L 0 0 0 0 % 0 0 0 0 d 0 0 0 0 d N " m`.
Please use other characters and try again.
Alternatively, enable Column Mapping to keep using these characters.

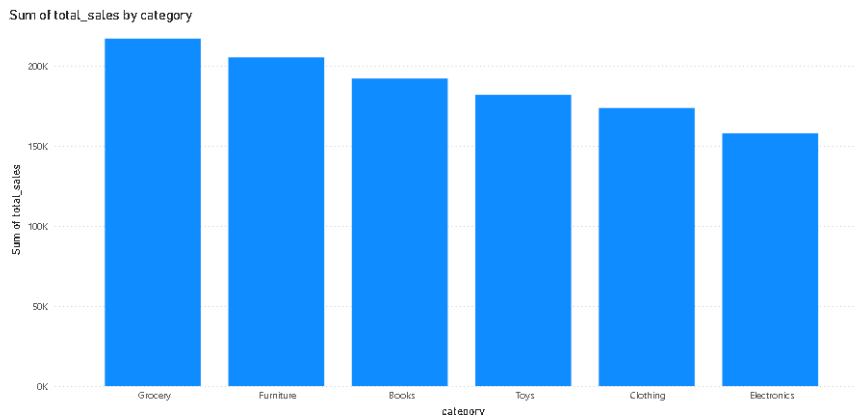
```

POWERBI LIVE DASHBOARD:



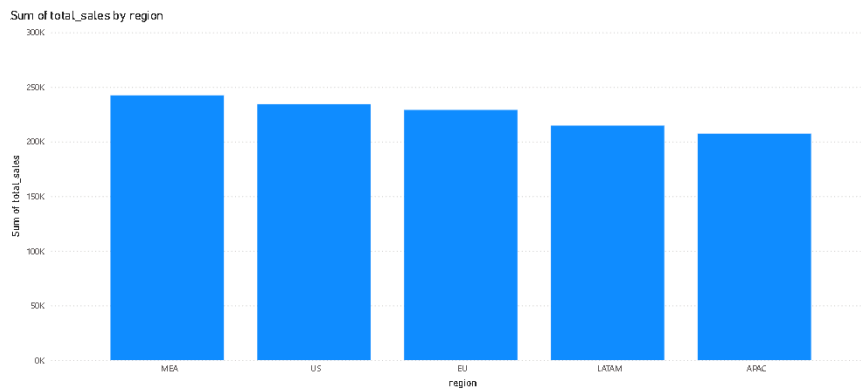
By Basab Saha

How much Sales 💰💰 from Each Category of Products 🧸

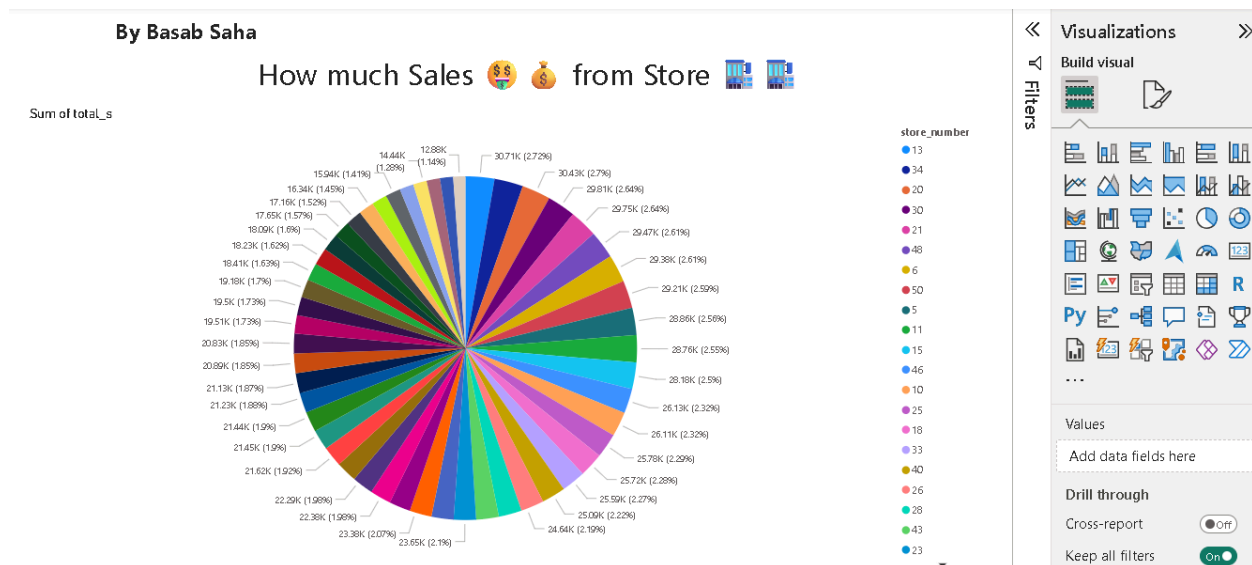


By Basab Saha

How much Sales 💰💰 from Each Region 🌐🌐



The screenshot displays the Tableau Desktop interface. On the left, the 'Visualizations' pane is open, showing a 'Build visual' section with a grid of chart types (bar, line, pie, etc.) and a 'Values' section with a search bar containing 'Add data fields here'. On the right, the 'Data' pane is visible, showing a list of fields with checkboxes for selection.



PERMISSIONS GIVEN TO DIFFERENT GROUP OF ENGINEERS :

Data Engineers Given : Modify and Select Permission to Bronze and Silver tables

Catalog Explorer > basab_catalog3 > all_delta >

bronze_orders

Open in a dashboard

Share

Create

Overview Sample Data Details **Permissions** History Lineage Insights Quality

Grant Revoke Privileges Inherited Type to filter by principal

Principal	Privilege	Object
data_engineers_basab	MODIFY	basab_catalog3.all_delta.bronze_...
data_engineers_basab	SELECT	basab_catalog3.all_delta.bronze_...

Data Analysts Given : Select Permission to Gold Aggregated tables

 **daily_region_sales**   

Open in a dashboard 

Share 

Create 

Overview

Sample Data

Details

Permissions

History

Lineage

Insights

Quality

Grant

Revoke

Privileges 

Inherited 

Type to filter by principal 

Principal	Privilege	Object
data_analysts_basab	SELECT	 basab_catalog3.all_delta.daily_re...